

Vigorimeter grip strength in CIDP: a responsive tool that rapidly measures the effect of IVIG – the ICE study

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Background and purpose: In a recent trial in patients with chronic inflammatory demyelinating polyradiculoneuropathy (CIDP), the ICE study, grip strength measurement captured significantly more improvement in patients receiving immune globulin (IGIV-C) intravenously than in those receiving placebo.

Methods: We conducted a systematic analysis to determine the sensitivity of grip strength as an indicator of meaningful clinical changes in CIDP.

Results: A randomized double-blind trial was undertaken in 117 CIDP patients who received IGIV-C or placebo every 3 weeks for up to 24 weeks. Grip strength and inflammatory neuropathy cause and treatment (INCAT) disability scores were assessed at each visit, and the responsiveness of each scale was compared. A minimum clinically important difference cut-off value for grip strength (>8 kPa) and INCAT score (>1 point) was applied to assess the proportion of responders to IGIV-C versus placebo. This analysis showed that grip strength demonstrated significant improvement earlier (as early as day 16) than the INCAT disability scale in patients receiving IGIV-C compared with placebo. A significantly higher proportion of improvers were seen in the IGIV-C group (37.5–50.9%) than in the placebo group (21.1–25.9%) for grip strength at day 16, week 3, week 6 and the end of the first period. Also, grip strength showed within the first 6 weeks in the placebo group significantly more patients with a clinically meaningful deterioration (>8 kPa), compared with the INCAT (>1-point deterioration) findings.

Conclusions: Grip strength can be considered a sensitive tool for assessing clinically relevant changes in patients with CIDP. Its use in daily practice is suggested.

Introduction

At a consensus meeting on outcome measures in inflammatory neuropathies, the level of disability was chosen to be the primary measure of assessing efficacy in these patients [1]. Therefore, the adjusted inflammatory neuropathy cause and treatment (INCAT) disability score – a 10-point ordinal measure capturing problems in daily arm and leg activities and mobility – was chosen as the primary outcome in the largest trial

ever performed in patients with chronic inflammatory demyelinating polyradiculoneuropathy (CIDP), the ICE study [2,3].

This disability measure has demonstrated statistically significant differences in favour of those patients being treated intravenously with human immune globulin, 10% caprylate/chromatography purified [immune globulin (IGIV-C)] compared with those receiving placebo [3]. However, whilst the adjusted INCAT disability score is a very useful tool with which to judge treatment response, it is not the typical measurement employed by neurologists in daily practice. By contrast, grip strength is measured using the

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Vigorimeter, which is a simple tool that is easy to apply and provides an instantaneous result [4]. Grip strength using a hand-held dynamometer, or an equivalent air-pressure tool such as the Martin Vigorimeter, is relatively objective and less susceptible to bias, but it is limited as an outcome measure because impairment is assessed solely for the hands [5]. In the ICE trial, grip strength also demonstrated a statistically significant improvement in favour of IGIV-C over placebo [3]. However, a systematic investigation has not been performed determining the sensitivity of grip strength versus the INCAT disability score as indicators of relevant clinical changes in CIDP. The purpose of the study reported here was to systematically examine the sensitivity of grip strength measurement and establish whether changes measured by this tool during a period of IGIV-C administration versus placebo are clinically meaningful and thus indicative of a tool that could be used also in daily practice as a good indicator of improvement or worsening in patients with CIDP.

Methods

Study design

For the purposes of this study, the ICE trial first-period data for the INCAT disability score and dominant hand grip strength scores were used [3]. Complete details of the patient population, study design and methodology are described in the publication of the primary efficacy analysis [3]. The study design (ClinicalTrials.gov identifier: NCT00220740) was approved by a regional review board. Briefly, the ICE study was a randomized, double-blind, placebo-controlled, response-conditional (rescue) trial and included a first period and a crossover (rescue) period [3]. Eligible participants were randomized to receive either IGIV-C (Gamunex[®], Grifols Therapeutics, Inc., formally Talecris Biotherapeutics Inc., Research Triangle Park, NC, USA) or placebo (0.1% albumin). Participants received a baseline loading dose of 2 g/kg (commencement of first or crossover periods) and then a maintenance infusion of 1 g/kg every 3 weeks for up to 24 weeks. An adjusted INCAT disability score was applied to assess functional disability prior to every subsequent infusion of study drug. Any participant with an adjusted INCAT disability score that worsened from baseline by ≥ 1 point at any time from day 16 until week 24 during the first period, or who had no improvement by week 6, was required to switch to rescue therapy during the crossover period.

Assessment tools

INCAT disability scale

This disability measure was applied to capture daily activities such as dressing the upper part of the body, doing and undoing buttons and zips, washing and brushing hair and handling coins. Each item is scored as being 'not affected', 'affected but not prevented' or 'prevented'. The leg scale measures problems with walking, taking into account the use of aids. The INCAT scale ranges from 0 (no signs of disability) to 10 (most severe disability score) [2]. For the purposes of the ICE study, the adjusted INCAT disability scale was used, meaning that a change in the upper limb part of this scale from 0 to 1 or *vice versa* was not considered to be clinically meaningful and was therefore not included in the analyses as being a responder.

Vigorimeter

The Vigorimeter (Martin, Tuttlingen, Germany) quantifies grip strength on a manometer that is connected to a middle-sized rubber bulb through rubber tubing (score range, 0–160 kPa) [4,5]. The average value of three assessments at each visit was used for the purposes of the current study.

Study procedure

Both the INCAT disability score and grip strength were assessed at each visit during the first period. Grip strength examination was performed under pre-defined standard conditions [6]. All participating centres were trained prior to the start of the study on how to assess the measures to ensure the consistency of application.

Statistics

Early response analyses

These analyses assessed whether grip strength as an easy and practical method could be used as a diagnostic tool in individual subjects after only one or two courses of treatment by examining statistical differences at the early visits between the two groups (IGIV-C and placebo). Grip strength data were compared with INCAT disability scale results.

Changes in global health

The assessment of global change in health used the answers to question number 2 of the Short Form 36 questionnaire (Medical Outcomes Trust, Boston, MA, USA; SF-36), which asks patients to compare their global health rating at study entry and at week 24 of the first period [7]. This question is not used in scoring the domains and summary measure of the SF-36.

The following question was asked to patients: ‘Compared to 1 year ago, how would you rate your health?’ Patient response options included five categories: ‘much better’, ‘somewhat better’, ‘about the same’, ‘somewhat worse’ or ‘much worse’.

Converting group response to individual response for grip strength using the minimum clinically important difference concept

Grip strength measure is a continuous variable, and the statistically significant group change does not necessarily imply a meaningful difference for individual patients. In the ICE study, a ≥ 1 -point improvement in the INCAT scale was used as the primary measure of a clinically meaningful change [2]. Grip strength findings can also be converted from a mean group response to an individual response assessment. For the individual response assessment, the grip strength measurement at each visit for each subject was evaluated against a cut-off value to determine whether there had been a clinically meaningful response (particularly examining the improvement in the IGIV-C group as well as the rate of deterioration [the so-called meaningful negative responders] in the placebo group). The cut-off value was based on the minimum clinically important difference (MCID) universal rule of $0.5 \times \text{SD}$ as was proposed by Sloan and colleagues [8]. In another study, we estimated the SD from the baseline measurement of all CIDP subjects in the ICE study, and MCID was 8 kPa for the Vigorimeter [9]. This MCID value was used to determine the proportion of patients in both treatment groups who were categorized as responders.

Correlation and agreement between INCAT disability scale responses and grip strength responses

The correlation between these two measures was assessed to determine whether grip strength could indirectly provide information on patients’ daily functionality, as has been suggested previously [5]. The agreement between the INCAT disability scale responses and the grip strength (dominant hand) responses was also analysed. Kappa coefficient was calculated for assessing the agreement. In addition, the proportion of subjects with clinically meaningful deterioration assessed by INCAT and grip strength was compared and tested using the McNemar’s test.

Statistical measures of responsiveness

INCAT disability scale and grip strength were measured at baseline, day 16, week 3, week 6 and every 3 weeks up to week 24. The data up to week 6 were utilized to assess which of these scales was more sensitive in detecting early responses. The standardized response

mean (SRM) and area under the curve (AUC) approaches were used to assess responsiveness [5].

For each scale, SRM is calculated as the mean change divided by the standard deviation in change scores. AUC is calculated using the linear trapezoidal rule

$$AUC_{0\text{-}week6} = \sum_{i=2}^4 \frac{(SRM_i + SRM_{i-1})(D_i - D_{i-1})}{2},$$

where SRM_i is the SRM at visit i and D_i is the number of days at visit i at baseline, day 16, week 3 and week 6.

Statistical description

Data were evaluated from the intention-to-treat population, defined as all randomized subjects. Data obtained in the first period were used for the purposes of this paper. A last observation carried forward approach was applied for the change from baseline variables. Analysis of covariance models were used for group response analyses, with treatment and region as fixed effects and baseline measure as covariate. For categorical variables, treatment differences were compared using chi-square tests. Correlation between INCAT disability scale values and grip strength scores was investigated using Pearson’s correlation test. For responsiveness assessment, the SRM was calculated as the mean change divided by the SD in change scores and had no unit. Subsequently, AUC was calculated for SRMs using the trapezoidal rule. The kappa statistic was applied to determine the agreement between the proportion of patients showing a response on both measures. All statistical analyses were performed using SAS® version 9.2 (SAS Institute, Cary, NC, USA).

Results

General characteristics

One hundred and seventeen individuals were randomized: 59 patients received IGIV-C and 58 patients received placebo. Baseline mean values for grip strength and INCAT disability scale were similar between the two treatment groups [3].

Early group response analyses

Statistically significant group response improvement differences were observed at week 6 for both measures (grip strength and INCAT disability scale), in favour of patients treated with IGIV-C. However, only grip strength (dominant hand) measured using the Vigorimeter achieved statistically significant differences at

earlier visits (day 16 and week 3) in favour of the IGIV-C group (Table 1). Grip strength (non-dominant hand) showed similar results where statistically significant differences ($P = 0.043$) were observed as early as week 3.

Table 1 Change from baseline to day 16, weeks 3, 6 and 24 (endpoint) of first treatment period for adjusted inflammatory neuropathy cause and treatment (INCAT) score and grip strength

Parameters	Visit/change from baseline	IGIV-C		Placebo		Treatment difference (95% CI)	P-value
		N	Mean $\pm$ SD	N	Mean $\pm$ SD		
Adjusted INCAT score	Day 16	57	-0.4 ± 1.0	57	-0.1 ± 0.7	$-0.31 [0.61, 0.00]$	0.051
	Week 3	57	-0.6 ± 1.1	51	-0.3 ± 0.8	$-0.22 [-0.55, 0.11]$	0.196
	Week 6	55	-1.0 ± 1.3	50	-0.4 ± 0.9	$-0.59 [-1.00, -0.17]$	0.006
	Endpoint	59	-1.1 ± 1.8	58	-0.3 ± 1.3	$-0.72 [-1.27, -0.18]$	0.010
Grip strength (dominant hand)	Day 16	56	7.9 ± 12.1	57	1.5 ± 14.7	$5.98 [1.04, 10.91]$	0.018
	Week 3	57	9.0 ± 11.8	51	2.0 ± 13.8	$5.72 [0.88, 10.56]$	0.021
	Week 6	54	11.9 ± 15.1	50	3.3 ± 12.7	$7.53 [2.21, 12.85]$	0.006
	Endpoint	57	13.2 ± 19.3	58	1.5 ± 15.6	$10.94 [4.65, 17.22]$	<0.001

Table 2 Individual clinically meaningful improvement response in grip strength (dominant hand) by treatment arm using the universal rule for the MCID ($0.5 \times \text{SD}$)

Visit	IGIV-C		Placebo		P-value
	N	Response (%)	N	Response (%)	
Day 16	56	37.5	57	21.1	0.055
Week 3	56	39.3	51	23.5	0.080
Week 6	54	55.6	50	24.0	0.001
Last measurement in first period	57	50.9	58	25.9	0.006

IGIV-C, Immune globulin; MCID, minimum clinically important difference; Response is defined as change in grip strength of at least 8 kPa.

Changes in global health

Figure 1 shows the relationship between the change in grip strength and the five categories of the global change in health as assessed using question number 2 of the SF-36. The figure shows that subjects with more improvement in grip strength tended to rate their improvement in global health better than those with less improvement in grip strength (Spearman's correlation coefficient: $r = 0.47$, $P < 0.001$).

Converting group response to individual response for grip strength using MCID

The cut-off value for grip strength was based on the MCID concept of $0.5 \times \text{SD}$ as the universal rule and was 8 kPa [8,9]. Table 2 shows the proportion of individual meaningful improvement responses using the MCID cut-off. There were more responders in

IGIV-C group than in the placebo group at each of the first three visits (day 16, week 3 and week 6). The results are similar to the proportion of individual responses using the adjusted INCAT score (data not shown).

Table 3 Clinically meaningful deterioration assessed by the inflammatory neuropathy cause and treatment (INCAT) disability score and grip strength in the placebo group

Visit	INCAT score significant negative response	Grip strength significant negative response N (%)		P-value
		No	Yes	
Day 16	No	44 (77.2)	9 (15.8)	0.003
(N = 57)	Yes	0	4 (7.0)	
Week 3	No	43 (84.3)	7 (13.7)	0.008
(N = 51)	Yes	0	1 (2.0)	
Week 6	No	38 (76.0)	8 (16.0)	0.058
(N = 50)	Yes	2 (4.0)	2 (4.0)	

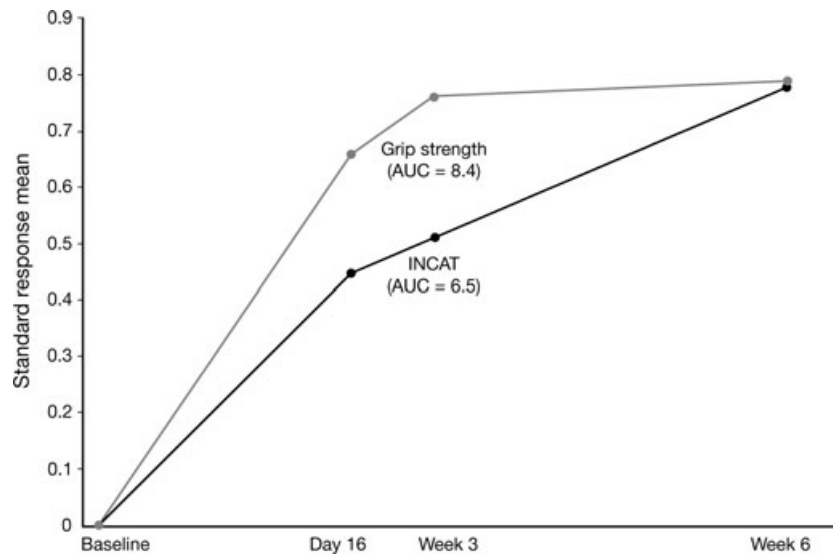
'Significant negative responder' was defined as a patient showing deterioration defined as (i) 1-point increase (=worsening) on the INCAT; (ii) 8-point deterioration on the Vigorimeter.

In the placebo group, a higher proportion of patients with a meaningful deterioration was captured by grip strength (negative responders showing >8 kPa reduction) within the first 6 weeks of the study when compared with the INCAT score findings (Table 3). At day 16, 13 patients (22.8%) were negative responders with the grip strength and only 4 (7%) with the INCAT disability score ($P = 0.003$); at week 3: grip strength; INCAT = 8 (15.7%) vs. 1 (2%) ($P = 0.008$), at week 6: grip strength; INCAT = 10 (20%) vs. 4 (8%) ($P = 0.058$) (Table 3).

Correlation and agreement between INCAT disability scale responses and grip strength responses

The correlation between the INCAT disability scale values and grip strength scores was highly significant

Figure 1 Change in grip strength (dominant hands) related to global health changes. Changes in global health were assessed using question number 2 of the SF-36 questionnaire: 'compared to 24 weeks ago, how would you rate your health?'



(at baseline: $r = 0.44$; at first-period endpoint: $r = 0.61$; $P < 0.001$). The results in Tables 3 and 4 show a fair agreement between INCAT disability scale responses and the grip strength score responses, with agreements ranging from 63.6% to 84.3%. However, there was some discordance, where INCAT changed by ≥ 1 point but grip strength improved by < 8 kPa, or grip strength improved by > 8 kPa but there was no improvement in INCAT disability score.

Statistical measure of responsiveness

Amongst the IGIV-C group, larger SRMs were observed at day 16 and week 3 for grip strength score than for INCAT disability score (Fig. 2). By week 6, both scales had similar SRMs (0.78 for INCAT disability score and 0.79 for grip strength). The AUC over the 6-week period was 8.4 for grip strength and 6.5 for INCAT disability score.

Table 4 Agreement between the inflammatory neuropathy cause and treatment (INCAT) disability score and grip strength responses

Visit	INCAT score response	Grip strength response N (%)		Kappa statistic
		No	Yes	
Day 16 ($N = 113$)	No	72 (63.7)	19 (16.8)	0.359
	Yes	8 (7.1)	14 (12.4)	
Week 3 ($N = 107$)	No	57 (53.3)	23 (21.5)	0.110
	Yes	16 (15.0)	11 (10.3)	
Week 6 ($N = 104$)	No	42 (40.4)	12 (11.5)	0.380
	Yes	20 (19.2)	30 (28.8)	

Discussion

This study provides data showing the clinical value of using the Martin Vigorimeter, an easily applicable tool for the assessment of grip strength in patients with CIDP. Statistically significant, clinically meaningful improvement in CIDP patients being treated with IGIV-C, compared with those given placebo, was captured using the Vigorimeter, and these changes were seen as early as day 16 of follow-up in the ICE study. By contrast, statistically significant changes were not seen with the INCAT disability score until week 6 (Table 1) [3]. Relevant changes in grip strength showed a good correlation with the patients' own judgement of change in their global health measured using question number 2 of the SF-36 (Fig. 1) [7]. Generally, individuals with an improvement in grip strength reported better improvement in global health and *vice versa*. The latter findings are considered a heuristic form of responsiveness [10]. In addition, the responsiveness assessed by the non-unit SRM and AUC indicated that grip strength is a more sensitive measure of response to treatment at early visits than the INCAT disability score. An extension of the responsiveness comparison between the two outcome measures beyond week 6 was limited because of the conditional crossover design of the study [2].

The concept of MCID has been thoroughly addressed in a recent study using the ICE trial data [9]. For both the INCAT disability scale and grip strength scores, higher proportions of responders were seen in the IGIV-C group than in the placebo group for the anchor-based and distribution-based MCID used. In the current study, a higher proportion of patients reached the cut-off MCID value of 8 kPa in

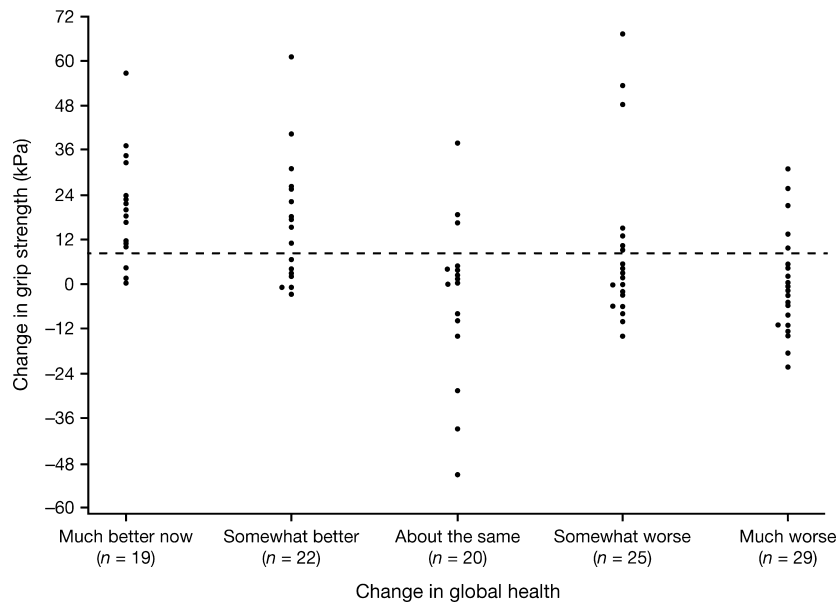


Figure 2 Standardized response mean (SRM) versus treatment duration for inflammatory neuropathy cause and treatment (INCAT) score and grip strength. SRM values are shown, together with corresponding area under the curve, for INCAT score and grip strength for subjects treated with IGIV-C as part of the early response comparison.

the IGIV-C group, thus showing a clinically meaningful improvement (Table 2). Hence, in addition to statistically significant comparison findings for the groups, these findings demonstrate higher response rates for both measures in favour of the IGIV-C group at the individual level.

The Martin Vigorimeter also demonstrated higher sensitivity at the other end of the spectrum of changes: this tool significantly captured as early as day 16 more patients in the placebo group showing a meaningful deterioration (MCID cut-off >8 kPa) when compared with the INCAT findings (Table 3). To our knowledge, these findings have not been published previously and could be used as an early indication for additional therapy if patients continue to deteriorate despite having received the first full IVIG dosage. However, future studies should focus on the impact of a second IVIG dosage in these patients and determine whether they respond more rapidly. In a recent study, it was demonstrated that some subjects respond (as determined by the INCAT score) to IVIG therapy after one infusion course, whilst other patients required two infusion courses to see a response [11]. This also holds true with the grip strength data, which showed the proportion of responses significantly increased from 37.5% at day 16 and 39.3% at week 3 to 55.6% at week 6 (Table 2). However, further studies are needed to explore the factors that might be predictive of needing more IVIG courses before a response to therapy is observed. These observations could also be of particular interest for the design of future maintenance trials in patients with CIDP.

All these findings indicate that, in addition to being simple, quick and easy to use and fulfilling the clinimetric requirements of being valid, reliable and responsive, the Vigorimeter detects responses quickly and is related to patients' own judgement of their clinical state (Fig. 1) [5]. Grip strength also demonstrated an acceptable correlation with the INCAT disability scale with moderate response agreement (Tables 3 and 4), which conforms previous reports [5,12,13]. However, there were also discordances between the two measures where INCAT score showed improvement without grip strength improvement and *vice versa* (Tables 3 and 4). This may be explained by some subjects having greater lower extremity disability changes that INCAT can detect, but grip strength cannot. Also, changes in grip strength (located at the impairment level of assessing outcome) do not always translate to a change in disability. Hence, the moderate agreement indicates that other factors exist that contribute to disability.

There are some methodological issues that should also be addressed. Although the Vigorimeter is easily applicable, fulfils all clinimetric requirements and correlates moderately with disability, it only assesses impairment distally and does not provide information on clinically important proximal muscle groups. Dynamometric devices have been suggested to overcome this deficit, making it possible to quantify possible proximal weakness [14,15]. Although seemingly more comprehensive in their applicability, these devices have a higher burden on patients and may even cause some discomfort [5]. In addition, as has been demonstrated for the Vigorimeter, no other grip

strength device has been subjected to a thorough examination to determine its scientific soundness in these illnesses. Although the use of the Vigorimeter in daily practice is suggested, physicians should also focus on other aspects that might contribute to problems in daily living and social functioning for patients with CIDP [16]. Despite these observations, grip strength, reflecting distal strength and upper limb function, has shown its usefulness as a prognostic indicator of clinical and functional recovery and is useful in monitoring the effect of treatment [12,13,17].

In conclusion, grip strength assessment with the Vigorimeter is a suitable outcome measure that is recommended for routine follow-up studies of patients with CIDP.

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Disclosure of conflict of interest

At the time of the work, Chunqin Deng and Kim Hanna were employed by, and owned shares in, Talecris Biotherapeutics Inc.; in addition, Chunqin Deng and Kim Hanna held stock options. Talecris Biotherapeutics Inc. was acquired by Grifols Inc., effective 1 June 2011. Chunqin Deng and Kim Hanna are currently employed by Grifols Inc. Norman Latov, Richard Hughes, Vera Bril, Marinos Dalakas, Peter Donofrio, Pieter Van Doorn, Hans-Peter Hartung and Ingemar Merkies received

honoraria for participation on the ICE Study Steering Committee. Richard Hughes also receives or has received consultancy fees from Baxter, Bristol-Myers Squibb, LFB, Novartis and Octapharma and serves on the board of GBS-CIDP Foundation International. Norman Latov is a recipient of a research grant from Talecris Biotherapeutics Inc., holds stock in Therapath LLC and received consultant's fees from Pfizer, Depomed, Baxter, Octapharma and Novartis. The neuromuscular research foundation at the Maastricht University Medical Centre received on behalf of Ingemar Merkies honoraria from Talecris Biotherapeutics Inc. for speaking at scientific symposia and for serving on a steering committee; Ingemar Merkies received funding for research from the GSB CIDP Foundation International. Vera Bril previously received consultancy fees and unrestricted educational grants for studies in myasthenia gravis (paid to her institution) from Talecris Biotherapeutics Inc. Marinos Dalakas has received a grant, consultancy fees and support for travel to meetings for study purposes from Talecris Biotherapeutics Inc., Baxter, and Octapharma; he also previously received payment for lectures from these same companies. Peter Donofrio has received a grant (paid to his institution), consultancy fees, fees for participation in review activities, and support for writing, medicines, equipment and administration from Talecris Biotherapeutics Inc.; he previously received consultancy fees from CSL Behring, Octapharma and Squid, along with payment for lectures and for the development of educational presentations from DIME. Pieter van Doorne and his institution were paid honoraria from Talecris Biotherapeutics Inc. for his membership of the ICE Study Steering Committee. Hans-Peter Hartung received a research grant from Baxter (paid to his institution), consultancy fees from Talecris Biotherapeutics Inc., Bayer Healthcare, and CSL Behring, and travel support from Talecris Biotherapeutics Inc. and CSL Behring. He previously received consultancy fees from Bayer Healthcare, Biogen Idec, Genzyme, Merck Serono, Novartis and Teva Sanofi Aventis and has also received grants from Bayer Healthcare, Biogen Idec, Merck Serono, and Teva (paid to his institution), as well as payments for lectures from these same companies. No disclosure for Els Vanhoutte.

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